WORKED SOLUTION

STEP II 2008, Question 1

Pure • Sequences

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q1

The key idea

The expressions $x^2 - y^2$ and $2xy$ are the real and imaginary parts of $(x + \mathrm{i}y)^2$. So if we write each point as a complex number $z_n = x_n + \mathrm{i}y_n$, the two-component rule becomes the single equation $z_{n+1} = z_n^2 + c$ with one complex constant c. Then "constant" and "period 2" become quadratic equations in one complex unknown, which we can solve with the quadratic formula.

Solution

Put $z_n = x_n + \mathrm{i}y_n$ and $c = a + (b+2)\mathrm{i}$. Since

$$z_n^2 + c = (x_n^2 - y_n^2 + a) + \mathrm{i}\,(2x_n y_n + b + 2),$$

and two complex numbers are equal exactly when their real and imaginary parts agree, the rule is equivalent to

$$z_{n+1} = z_n^2 + c.$$

Because a, b, x_n and y_n are real, every point (x, y) corresponds to exactly one complex number and vice versa. So nothing is lost by working with z.

Part (i)

With $a = 1$, $b = -1$ we have $c = 1 + \mathrm{i}$. The sequence is constant exactly when $z_2 = z_1$, since then every later step repeats the same calculation. So we need

$$z^2 - z + (1 + \mathrm{i}) = 0, \qquad z = \frac{1 \pm \sqrt{1 - 4(1+\mathrm{i})}}{2} = \frac{1 \pm \sqrt{-3 - 4\mathrm{i}}}{2}.$$

To find a square root of $-3 - 4\mathrm{i}$, look for $(p + q\mathrm{i})^2 = -3 - 4\mathrm{i}$ with p, q real: $p^2 - q^2 = -3$ and $pq = -2$. Then $p = 1$, $q = -2$ works, since $(1 - 2\mathrm{i})^2 = 1 - 4\mathrm{i} - 4 = -3 - 4\mathrm{i}$. Hence

$$z = \frac{1 \pm (1 - 2\mathrm{i})}{2} = 1 - \mathrm{i} \quad \text{or} \quad \mathrm{i}.$$

So the sequence is constant exactly when

$$(x_1, y_1) = (1, -1) \quad \text{or} \quad (x_1, y_1) = (0, 1).$$

Directly: $(1,-1) \mapsto (1-1+1,\ -2+1) = (1,-1)$ and $(0,1) \mapsto (0-1+1,\ 0+1) = (0,1)$.

Part (ii)

Now $z_1 = -1 + \mathrm{i}$, so $z_1^2 = 1 - 2\mathrm{i} - 1 = -2\mathrm{i}$ and

$$z_2 = c - 2\mathrm{i}.$$

Write $w = z_2$. The sequence has period 2 when $z_3 = z_1$ but $z_2 \neq z_1$. Once $z_3 = z_1$, the rule makes the sequence repeat as $z_1, z_2, z_1, z_2, \ldots$. Using $c = w + 2\mathrm{i}$,

$$z_3 = w^2 + c = w^2 + w + 2\mathrm{i},$$

so $z_3 = z_1$ becomes

$$w^2 + w + 2\mathrm{i} = -1 + \mathrm{i} \quad \Longleftrightarrow \quad w^2 + w + (1 + \mathrm{i}) = 0.$$

Ask yourself Why choose $w = z_2$ as the unknown, and not c?

The condition $z_3 = z_1$ is a quadratic in c either way, but in terms of w the constant sequence $w = z_1$ must be one root. So we already know one root of the quadratic, and the other is the one we want. Indeed $w = -1 + \mathrm{i}$ satisfies it: $(-1+\mathrm{i})^2 + (-1+\mathrm{i}) + 1 + \mathrm{i} = -2\mathrm{i} + 2\mathrm{i} = 0$. The roots sum to -1, so the other root is $w = -\mathrm{i}$. (The quadratic formula gives the same: $w = \frac{1}{2}(-1 \pm \sqrt{-3-4\mathrm{i}}) = \frac{1}{2}(-1 \pm (1-2\mathrm{i}))$, the same square root as in (i).)

- $w = -1 + \mathrm{i} = z_1$ gives $c = -1 + 3\mathrm{i}$, that is $a = -1$, $b = 1$. Here $z_2 = z_1$ and the sequence is constant, so its period is 1, not 2.
- $w = -\mathrm{i}$ gives $c = \mathrm{i}$, that is $a = 0$ and $b + 2 = 1$.

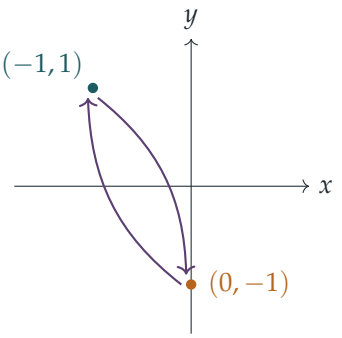
So the sequence has period 2 exactly when

$$a = 0, \qquad b = -1.$$

Check in coordinates: with $a = 0$, $b = -1$,

$$(-1, 1) \mapsto (1 - 1 + 0,\ -2 + 1) = (0, -1) \mapsto (0 - 1 + 0,\ 0 + 1) = (-1, 1),$$

and the two points are different, so the period is exactly 2.



Checks

- Iterating the original coordinate rule shows that $(1,-1)$ and $(0,1)$ are fixed when $a=1$, $b=-1$, and that $(-1,1)\to(0,-1)\to(-1,1)$ when $a=0$, $b=-1$.
- The square root $1-2\mathrm{i}$ can be checked by squaring it, and so can the factorisation $w^2+w+1+\mathrm{i}=(w+1-\mathrm{i})(w+\mathrm{i})$: the product of the roots is $(-1+\mathrm{i})(-\mathrm{i})=1+\mathrm{i}$.
- A numerical root search over random starting values found no other fixed points in (i) and no other (a,b) with $z_3=z_1$ in (ii) apart from $(-1,1)$, which gives the constant sequence.

Common slips

- Forgetting the "+2" in the second component, so taking $c=a+b\mathrm{i}$ in place of $a+(b+2)\mathrm{i}$.
- Giving $a=-1$, $b=1$ as a second answer to (ii). That sequence is constant, so its period is 1.
- Working only in real coordinates and getting lost in simultaneous quadratics. This works, but the complex form is much shorter.

Beyond STEP

The map $z\mapsto z^2+c$ is the most famous example in complex dynamics. The Mandelbrot set is the set of c for which the orbit of 0 stays bounded, and its two largest pieces, the main cardioid and the disc to its left, consist of the c for which the map has an attracting fixed point or an attracting 2-cycle. Here, too, the equation for 2-cycles had a factor coming from the fixed points: in general $(z^2+c)^2+c-z=(z^2-z+c)(z^2+z+c+1)$. For a readable introduction to the geometry behind this, see Needham, *Visual Complex Analysis*.

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